

Geolocation Challenge

Problem Statement & Rules

AI Club Hackathon Vertical, IIT Madras

August 14, 2026

At a Glance

- **Task:** Predict location (latitude, longitude) *and* a confidence radius for unlabeled photos, given constraints on runtime compute.
- **Platform:** Kaggle -[Competition Link](#)
- **Compute Restrictions:** Free-tier only (Colab T4 / Kaggle P100/T4). No paid/rental cloud compute allowed, please stick to your local or the above
- **Two test sets:** one live from day one, a second (larger) one revealed only towards close of the hackathon. More details follow.
- **Every team must submit:** a fully executed notebook and a short approach write-up - these are also judged, in addition to your leaderboard score.

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1 Objective

If you’ve ever played [Geoguessr](#), you know exactly the kind of fun that’s gonna unfold in the coming days(with a few twists, ofc)! For those of you still in the dark, here’s a quick 101:

In the original game, you’re given a Google Maps street-view type interactive interface, and you move around(or not if you’re a pro) to find clues that help you locate it on a map. Scoring is simple, the closer you are to the real point, the more you get. This challenge asks of you a model that plays it competitively, at scale, and without cheating(on images).

Problem Statement

Produce, at the end of the challenge (design/train/finetune), a lightweight, efficient computer vision model capable of geolocating unlabeled street-level and landscape imagery- **without** querying any external API, reverse-image search engine, or mapping/geocoding service at inference time.

For every image in the test set, your model must output:

1. A predicted **coordinate**: latitude and longitude.
2. A predicted **radius** (in kilometers): how confident you are that the true location falls within that distance of your predicted point.

How you get there is up to you. You may treat this as continuous coordinate regression, as classification into discrete geographic cells with an uncertainty estimate, or as a hybrid of the two - the scoring mechanics (Section 5) are designed to reward accurate, well-calibrated answers regardless of the internal representation you choose. What is fixed is the constraint that matters most: your solution must be able to run end-to-end on a single free-tier GPU, entirely offline, on hardware you do not control at evaluation time.

2 Event Timeline and Platform

The competition is hosted as a [Kaggle competition linked here](#). You can submit predictions as CSV files through Kaggle’s standard submission flow; a custom evaluation metric scores your submission and populates the leaderboard.

Milestone	Date / Window
Competition launch	August 14, 2026
Submission deadline	August 19, 2026 EOD
Hidden Test Set release	Closing hours
Hidden Test Set submission window	Will be communicated
Notebook + write-up submission	Shortly after final deadline, form link

Note that *two* hidden test sets are used in this competition, released at different times. Both contribute to your final score.

3 Datasets

3.1 Provided Training Data

You are provided with a curated dataset of geotagged images together with per-image metadata: latitude, longitude, and ISO country code(which can be derived from the other info). Images are provided as standard JPEG files; metadata is provided as a CSV keyed by image filename. The training dataset can be found at – [Dataset Link](#).

3.2 External Data

You are free to source and use any additional publicly available dataset to train your model. There is no restriction on the scale or origin of your training data, subject to two conditions:

- You must not use the actual hidden evaluation images - or any external source of their ground-truth coordinates - anywhere in your training pipeline, even inadvertently.
- Any external dataset used must be disclosed in your write-up (Section 4).

You are responsible for ensuring you have the rights to use any external dataset you bring in.

4 Rules and Constraints

4.1 Compute Constraints

- Training and experimentation must be done on **free-tier resources only** - Google Colab (T4 GPU) or Kaggle Notebooks/Kernels or on your local PC.
- No paid cloud compute of any kind (dedicated GPU rentals, paid Colab Pro tiers used for compute advantage, etc.) is permitted.
- Your submitted training notebook (Section 4.7) must make this verifiable.

4.2 Foundation Models and the Zero-Shot Rule

You may use any pretrained foundation model (ResNet, ViT, CLIP, DINO, SigLIP, and similar) as a backbone. However, you must **fine-tune or train new parameters specifically for this**

task on top of that backbone. Using a foundation model purely zero-shot - with no task-specific training of any kind - is not a valid submission.

4.3 Offline Inference is Mandatory

All inference against both hidden test sets must run **100% offline**, with no network access of any kind at inference time.

Concretely:

- **Externally hosted commercial APIs or models** (including but not limited to GPT-4o, Gemini, and Claude) are **banned** from any part of your inference pipeline.
- This applies specifically to *generating or assisting in generating a location guess*. Using such tools for general coding assistance, debugging, or write-up drafting is fine.
- All model weights your submission depends on must be bundled with your notebook or loaded from a source that does not require network access at inference time. Nothing may be downloaded on the fly during scoring.

4.4 Vision-Language Models (VLMs): Here are some constraints

Local, open-weight vision-language models (Qwen-VL, LLaVA, PaliGemma, and similar) are permitted on free-tier hardware, but only in a specific way:

- **Allowed:** Take a local VLM, *freeze* its vision-encoder backbone (e.g. the SigLIP or CLIP encoder inside it), *discard* the language-generation head entirely, and train a new classification or regression layer on top of the resulting visual embeddings to predict geocells and/or coordinates with a confidence radius.
- **Not allowed:** Loading a VLM in its native form and simply prompting it with text to have it output a location guess directly. This counts as a zero-shot violation (Section 4.2) regardless of any fine-tuning elsewhere in your pipeline - the language-generation head must not be the thing producing your answer.

4.5 Test Set 1 - Submission Rules

Standard Kaggle competition submission format applies for the duration of the competition. A daily submission cap of 15 submissions per team is enforced to discourage brute-force probing of the leaderboard.

4.6 Hidden Test Set - Time-Boxed Submission Window

Hidden Test Set 2 will be released for a limited period of time after the competition completes, as another Kaggle competition. Please ensure you submit this in order to see your final leaderboard standings, as well as to be eligible for evaluation.

4.7 Required Deliverables

Every team, regardless of leaderboard placement, must submit:

1. Your final prediction CSV for each test set (submitted through Kaggle in the required format.

2. A **complete, fully executed** notebook (all cells run, with outputs visible):
 - All feature engineering and data augmentation steps used(if any).
 - The full training pipeline for your best-performing model.
 - Any other techniques or approaches you explored along the way, even ones you abandoned.
 - The code used to generate the evaluation CSV submitted.
3. A short written approach summary, submitted separately via the Google Form link, describing your methodology, the key decisions you made, and what you tried.

Submissions that cannot be reproduced from the submitted notebook, or those which require network access for inference are not allowed.

5 Scoring Mechanics and Evaluation

5.1 What You Submit Per Image

For every test image, you submit a predicted **coordinate** (latitude, longitude) and a predicted **radius** in kilometers. Look up the actual name of these columns in the sample *submission.csv* and *match*.

5.2 Per-Image Score

Each prediction is scored on three components:

- **Distance score:** a smooth, continuously decaying score based on the great-circle (Haversine) distance between your predicted point and the true location. Closer guesses score higher; there is no hard distance cutoff, but let me tell you: exponentials fall off pretty fast.
- **Calibration bonus/penalty:** your claimed radius is checked against your actual error. A tight radius that correctly covers the true location is rewarded more than a wide, “safe” radius that also covers it. A tight radius that *misses* is penalized more than a wide radius that misses – confidently wrong is worse than honestly unsure.
- **Country bonus:** an additional bonus if your predicted center falls within the correct country *and* your claimed radius is reasonably tight.

Your score on a given test set is the **median** of your per-image scores(not mean, because we’re nice people).

5.3 Two Test Sets

- **Test Set 1** is released alongside the competition launch and is scored live throughout. You should assume it is comparatively modest in size.
- **Test Set 2** is released only as described above. You should assume, and it is, **considerably larger** than Test Set 1.
- Your **final score** is a weighted combination of your performance on both test sets.

5.4 Evaluation

Final rankings are **not** decided by the leaderboard number alone. Your submitted notebook and write-up (Section 4.7) will be reviewed, and we will weigh:

- Depth of technical understanding demonstrated in your approach.
- Soundness and creativity of your methodology.
- Code quality, rigor, and reproducibility.

Please keep all your code reproducible, using seeds wherever possible

6 Hosting and Submission Format

6.1 Platform

The competition is hosted on Kaggle. The primary competition URL is given at the start. Towards the end, a secondary URL will be shared for the hidden test set.

6.2 Submission File Format

Each submission is a CSV file with exactly the following columns, one row per test image:

Column	Description
image_id	Identifier matching the test image filename
pred_lat	Predicted latitude (decimal degrees)
pred_lon	Predicted longitude (decimal degrees)
pred_radius_km	Predicted confidence radius, in kilometers

Submit one such file for Test Set 1 during the main competition window, and a second for Test Set 2 during its window.

All the best!